Chemical Energy

CCC-624

How do we get energy from the food?

Unlike plants, which can get energy directly from sunlight, animals (including humans) must get energy from the food they eat.

Process same like burning wood but different energy

- When the cells of your body burn fuel (sugar), produce C02
- During **cellular respiration** animal cells combine oxygen with food molecules to release energy to live and function.
- Remember that cellular respiration produces carbon dioxide as a waste product.
- Animals use energy to grow, reproduce, and to function. They release the carbon dioxide into the air as a waste product.
- Plants help animals and animals help plants.

History

- It wasn't until the 1850s that scientists realized the Earth is a closed system-nothing comes in or goes out. Elements in our environment have cycles. They are used again and again. Recycling is a natural part of our existence. That's one reason we need to be careful and keep our environment clean and healthy

Respiration and Photosynthesis

- Oxygen + Glucose $\longrightarrow$Energy + Carbon dioxide + water

- Light + Carbon Dioxide +water $\longrightarrow$ Oxygen + Glucose

Introduction

- Perhaps the most convenient form in which to *store* energy is chemical energy.

- Internal Energy U, sum of KE, PE, ChE, EE, Nuclear Energy and all other energise possessed by atoms and molecules of the system.

- The foods we eat, combined with the oxygen we breathe, store energy that our bodies extract and convert into mechanical and thermal energy.

Examples

The fuels that we burn in our automobiles, furnaces, and campfires also store energy in chemical form.

Batteries store chemical energy as well, for later retrieval in the form of electrical energy.

Chemical reaxtion

- A chemical reaction is simply a rearrangement of the way that atoms are connected together to form molecules.

- $H_2 + 1/2O_2 \rightarrow H_2O$
- Initially, the hydrogen atoms are connected together in pairs to form H_2 molecules
- In practice, the other half of the O_2 will usually combine with another H_2, to form a second water molecule.

Contd.

- Many chemical reactions, especially those that consume oxygen, give off energy.
- Above one as an example half a billionth of a billionth of a joule,

- If the reaction occurs by ordinary combustion (burning), then the energy is given off as thermal energy, but sometimes the energy released can take on other forms, such as electrical or even mechanical energy.

Clearly we need to deal with large numbers, why?

- **Avogadro's number** 6.02×10^{23}.

- It happens to equal the number of hydrogen atoms in one gram of hydrogen. A mole of of hydrogen atom has 1 gm.

A mole of hydrogen atoms has a mass of one gram, so a mole of H2 molecules has a mass of two grams.

Hydrogen is the simplest and lightest element. The other two

elements that **commonly occur in chemical fuels** are also among the lightest: carbon, which is 12 times heavier (per atom) than hydrogen, and oxygen, which is 16 times heavier than hydrogen.

Energy Release

- When one mole of hydrogen molecules (two grams) combines with half a mole of oxygen molecules (16 grams) to form one mole of water molecules (18 grams), the total energy released is greater: 286,000 joules.)

- burning a kilogram of hydrogen gas releases how much energy?

solution

- Since there are 500 moles of hydrogen gas in a kilogram, this means that burning a kilogram of hydrogen gas releases 500 times as much energy, or 121 million joules

Fossil Fuels

Hydrogen gas is rather uncommon on earth, so it is not widely used as a fuel.

Instead, we tend to burn carbon-containing fuels, the remains of organic material, mostly plants, that have been buried, compressed, and chemically altered over hundreds of millions of years.

The most common of these “fossil fuels” is coal, a complex mineral composed mostly of carbon, with small amounts of hydrogen, oxygen, and other elements mixed in.

Coal burning

- To a first approximation, we can pretend that coal is **pure carbon**, in the form of graphite. (The other familiar form of pure carbon is diamond, usually considered much too valuable to burn as a fuel!) The combustion of graphite results in the gas carbon dioxide, according to the reaction equation

Coal Burning

- $C + O_2 \longrightarrow CO_2$.

- For each mole of graphite burned, or equivalently, each mole of carbon dioxide produced, this reaction yields 394,000 J of energy. Amole of graphite is 12 grams, so a kilogram of graphite contains 1000/12 = 83 moles. Burning one kilogram of graphite therefore yields 83×394, 000 J, or 33 MJ. High-grade coal generally yields

little less energy, roughly 29 MJ per kilogram.

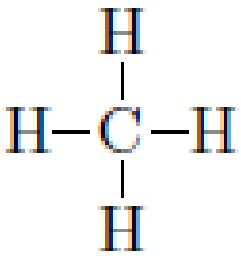
Other Fossil Fuels

- The other important fossil fuels are oil (petroleum) and natural gas, both of which are composed mostly of **hydrocarbons**: molecules containing both hydrogen and carbon.

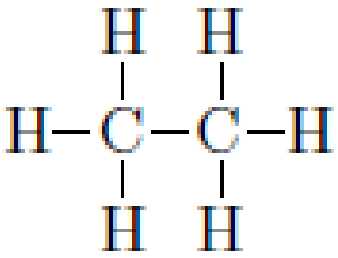
- The simplest and most common hydrocarbons are the so-called **alkanes**, in which each carbon atom is linked to four other atoms, each of which canbe either another carbon or a hydrogen.

Alkanes

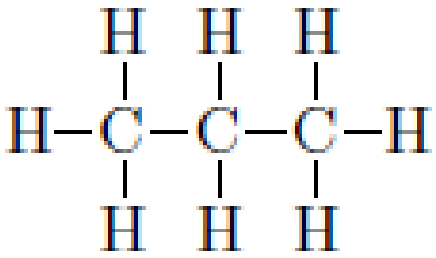
- The simplest **alkane** is methane, CH4, the principal component of natural gas. Next-simplest is ethane (C2H6), then propane (C3H8), then butane (C4H10), then pentane, hexane, heptane, octane, nonane, decane, and so on, where each molecule in the list is bigger than the previous one by a single carbon atom and two hydrogens.



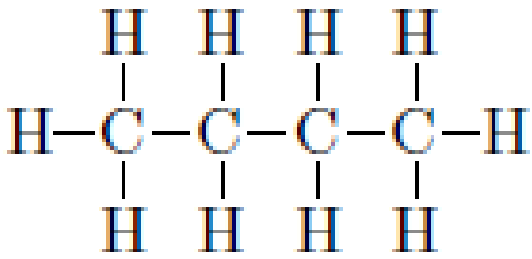
Methane

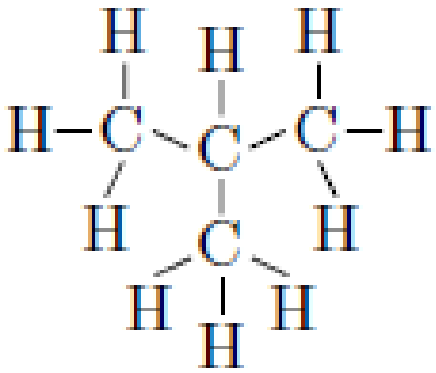


Ethane



Propane





Butane (two varieties)

- Each carbon atom is linked to four other atoms, while each hydrogen is linked only to a single (carbon) atom. Given these rules, you can draw similar diagrams for larger alkane molecules(pentane, hexane, heptane, octane, and so on).

Hydrocarbon combustion

- Because hydrocarbon molecules contain both hydrogen and carbon, their combustion in oxygen yields **both water and carbon dioxide** (as in reaction 4.3). You can figure out how much of each simply by requiring that the number of atoms of each type “balance” on both sides of the reaction equation.

Combustion of methane

Consider the combustion of methane, for instance. Since methane contains a single carbon atom, its combustion must produce a single CO_2 molecule. Meanwhile, to account for the four hydrogen atoms in methane, we must end up with two H_2O molecules. Finally, the CO_2 and the two H_2O's contain a total of four oxygen atoms, so the number of O_2 *molecules* that go into the reaction must be 2. The full reaction equation is therefore

$$CH_4 + 2O_2 \longrightarrow CO_2 + 2H_2O. \tag{4.4}$$

The end-products will always be water and carbon dioxide.

Energy produced by combustion

- The energy given off by the combustion of one mole of methane turns out to be 802 kJ. From this information, you should be able to show that the combustion of one *kilogram* of methane releases 50 MJ.

- Heavier hydrocarbons generally yield more energy per mole, but approximately the same energy on a per-kilogram basis.

- Gasoline, a mixture of hexane, heptane, octane, and various other hydrocarbons, yields about 44 MJ per kilogram,

Energy from Food

- Although fossil hydrocarbons are convenient chemical fuels, our bodies cannot digest them. Instead we obtain our energy from the more complicated molecules found in living plants and animals: sugars, fats, and proteins.

Example

- The “simple” sugars, such as glucose, have the chemical formula C6H12O6. The balanced reaction equation for “combustion” of glucose (whether by literal burning or the more subtle processes of metabolism) is

$$C_6H_{12}O_6 + 6O_2 \longrightarrow 6CO_2 + 6H_2O.$$

Energy perspective

- For one mole of glucose, this reaction releases 2.80 MJ of energy (assuming now that the water ends up in liquid form). That translates to 15.6 kJ, or 3.7 kcal, per gram. Dieticians normally just round this number to 4 kcal/gram, the “caloric content” of sugar. Sucrose, or table sugar, is basically just two simple sugar molecules linked together, while starch is a long chain of many simple sugar links. All such molecules are referred to as **carbohydrates**.

Point to be Noted

- Notice that carbohydrates yield considerably less energy than hydrocarbons, on a per-gram basis. This is because they already contain quite a bit of oxygen, so there is less opportunity for them to combine with oxygen to release energy. In a sense, they are already “partially burned.”

Contd.

- Fat molecules, on the other hand, are made almost entirely of carbon and hydrogen, with very little oxygen. Their metabolism in our bodies yields approximately 9 kcal/gram, more than twice the energy released from carbohydrates. Our bodies also produce fats as a way of storing energy for later use.

Contd.

- Proteins are very complex molecules containing considerable nitrogen in addition to carbon, hydrogen, and oxygen. They serve a variety of nutritional needs, but our bodies can metabolize them for energy when necessary, extracting approximately 4 kcal/gram, the same as from carbohydrates.

Exercise

- **Exercise 4.2.** What is the mass of one mole of ethyl alcohol (C_2H_5OH) molecules?

Exercise

- **Exercise 4.3.** In analogy with Figure 4.1, draw the structure of pentane, the alkane containing five carbon atoms. Try to draw at least two different varieties, analogous to the two varieties of butane. What is the chemical formula for pentane?

Exercise

- **Exercise 4.4.** Write down the reaction equation for the combustion of ethane, and balance it with the correct coefficients.

- Write down the reaction equation for the combustion of propane, and balance it with the correct coefficients.

Exercise

- **Exercise 4.7.** The combustion of one mole of propane ((C3H8) releases 2040 kJ of energy. How much energy is released by the combustion of one *kilogram* of propane? (Hint: First compute the mass in grams of one mole of propane.)

- **Exercise 4.8.** In a previous exercise you estimated the kinetic energy of a typical car traveling at highway speed. How much gasoline would be needed to provide this amount of energy, if we unrealistically assume 100% efficiency in the conversion of chemical to kinetic energy?